Performance Estimation Problems

A Tutorial – Session 1 / 3

March 2026

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March 16, 2026

Outline

1. Introduction & Motivation
2. Reminders: Gradient Descent and Convergence Proof
3. Performance Estimation – Squared Distance
4. Performance Estimation – Function Value
5. The Generic PEP Framework & PEPit
6. Minimal Working Code in PEPit
7. Primal Feasible Points as Counter-Examples
8. Hands-on: Counter-Examples & Dimension Reduction
9. Technical Leftover Details: Interpolation, Homogeneity, Matrices
10. Dual and Proofs
11. Conclusion

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Performance Estimation = computer-aided techniques for proofs in first-order optimization

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→ **Take away**: for most first-order algorithms and simple function classes, we can get the tightest convergence rate numerically.

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✘ Technical ? Simple to adapt?

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Setting: first-order optimization

Minimize $f : \mathbb{R}^d \to \mathbb{R}$:

$$x_\star = \arg\min_{x \in \mathbb{R}^d} f(x),$$

where $f \in \mathcal{F}$ (encodes assumptions on the function class).

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Regularity conditions: Smoothness

Definition : L-Smoothness

A differentiable function f is L-smooth if ∇f is L-Lipschitz:

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Proposition : Smooth + Convex characterization

A differentiable f is L-smooth and convex if and only if for all x, y:

$$f(y) + \langle \nabla f(y), x - y \rangle \;\leq\; f(x) \;\leq\; f(y) + \langle \nabla f(y), x - y \rangle + \tfrac{L}{2}\|x - y\|^2.$$

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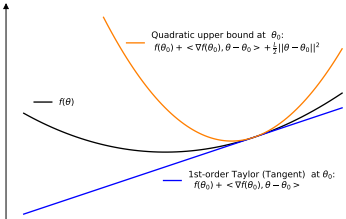

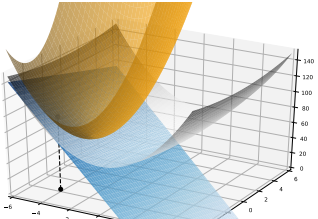

Observing f and ∇f *at* y provides information on the function *everywhere*.

Regularity conditions: Strong Convexity & the class $\mathcal{F}_{\mu,L}$

Definition : μ-Strong Convexity

f is μ-strongly convex if for all $x, y \in \mathbb{R}^d$:

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$\mathcal{F}_{\mu,L}$ denotes the set of μ-strongly convex and L-smooth functions on $\mathbb{R}^d$. (We allow $\mu = 0$ for convex-only.)

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Proposition : Co-coercivity (key inequality for GD analysis)

If $f \in \mathcal{F}_{\mu,L}$, then for all x, y:

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \;\geq\; \frac{\mu L}{\mu + L}\|x - y\|^2 + \frac{1}{\mu + L}\|\nabla f(x) - \nabla f(y)\|^2.$$

Gradient Descent: convergence rate

Theorem : Convergence of GD on $\mathcal{F}_{\mu,L}$

Let $f \in \mathcal{F}_{\mu,L}$ with $0 \leq \mu \leq L$.

GD with step size $\eta = \dfrac{2}{\mu + L}$ satisfies, after t iterations:

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→ The rate $\left(\dfrac{L-\mu}{L+\mu}\right)^2 \approx 1 - \dfrac{2\mu}{L}$ for $\kappa = L/\mu \gg 1$.

Proof of convergence (sketch)

Step 1. Expand the squared distance:

$$\|x_{t+1} - x_\star\|^2 = \|x_t - \eta \nabla f(x_t) - x_\star\|^2 = \|x_t - x_\star\|^2 - 2\eta \langle x_t - x_\star, \nabla f(x_t) \rangle + \eta^2 \|\nabla f(x_t)\|^2.$$

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Step 2. Apply co-coercivity (with $\nabla f(x_\star) = 0$):

$$\langle \nabla f(x_t),\, x_t - x_\star\rangle \;\geq\; \frac{\mu L}{\mu + L}\|x_t - x_\star\|^2 + \frac{1}{\mu + L}\|\nabla f(x_t)\|^2.$$

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$$\Rightarrow \text{By induction: } \|x_t - x_\star\|^2 \leq \left(\frac{L - \mu}{L + \mu}\right)^{2t}\|x_0 - x_\star\|^2.$$

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EXAMPLE ONE

—

Primal PEP for squared distance

Crash course on Performance Estimation Problems

$\rightarrow$ **Performance Estimation problems**: rethinking proofs of first-order optimization [1].

What is a worst-case guarantee (WCG)?

Example:

$$\forall f \in \mathcal{F}_{\mu,L}, \quad \text{for } x_1 = x_0 - L^{-1}\nabla f(x_0): \quad \|x_1 - x_\star\|^2 \leq \tau \|x_0 - x_\star\|^2$$

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Crash course on Performance Estimation Problems

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Crash course on Performance Estimation Problems

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Key insight The problem has been lifted to a semidefinite program (SDP) – a convex problem!

What did we just do? – Summary

$$\forall f \in \mathcal{F}_{\mu,L}, \quad \text{for } x_1 = x_0 - L^{-1}\nabla f(x_0): \quad \|x_1 - x_\star\|^2 \leq \tau \|x_0 - x_\star\|^2$$

Step 1. Write the tightest τ as a supremum:

$$\tau^* = \sup_{\substack{f \in \mathcal{F}_{\mu,L},\ x_0, x_\star \\ x_1 = x_0 - L^{-1}\nabla f(x_0),\ x_\star = \arg\min f}} \frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2}$$

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Steps 2–3. Reduce to a finite dimensional problem (no more f):

- Sample $g_0 = \nabla f(x_0)$; set $x_\star = 0$ and $\|x_0\|^2 = 1$ by homogeneity.
- Replace $\exists f \in \mathcal{F}_{\mu,L}$ by **interpolation conditions** (Taylor et al. '17).

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Key point: the problem is now finite-dimensional (variables $x_0, x_1, g_0 \in \mathbb{R}^d$) and quadratic in (x_0, g_0).
$\rightsquigarrow$ One more step: lift to a convex SDP.

Step 4: lifting to a semidefinite program

Step 4. Every quadratic in (x_0, x_1, g_0) is **linear** in the Gram matrix $G = vv^\top$, $v = (x_0, x_1, g_0)$:

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The PEP becomes a standard SDP:

$$\tau^* = \sup_{\substack{G \succcurlyeq 0 \\ \mathrm{Tr}(G\, C_{\mathrm{alg}}) = 0 \\ G_{11} = 1 \\ \mathrm{Tr}(G\, C_{\mathrm{class}}) \geq 0}} G_{22}$$

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Basic take-away

→ **Convex SDP**: solvable with any solver.

Interactive Session #1

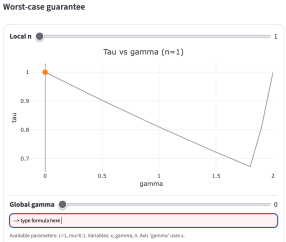
Exploring PEP numerically with the Streamlit UI

Activity: GD on L-smooth and μ-strongly convex functions — what is the worst-case rate for $\|x_1 - x_\star\|^2$?

1. Go to `pepitui-test.streamlit.app`
2. Select **Gradient Descent** (preimplemented), use *Customize* to set the performance metric to $\|x_1 - x_\star\|^2$ vs $\|x_0 - x_\star\|^2$
3. Set the function class to $\mathcal{F}_{\mu,L}$ (smooth **and** strongly convex)
4. Set $\mu = 0.1$, $L = 1$
5. Run `plot` and observe the worst-case value τ^*

$\Rightarrow$ **Try to guess the formula for** τ^* as a function of μ and L.

$\Rightarrow$ **Which step size gives the best rate?**



Outline

1. Introduction & Motivation
2. Reminders: Gradient Descent and Convergence Proof
3. Performance Estimation – Squared Distance
4. **Performance Estimation – Function Value**
5. The Generic PEP Framework & PEPit
6. Minimal Working Code in PEPit
7. Primal Feasible Points as Counter-Examples
8. Hands-on: Counter-Examples & Dimension Reduction
9. Technical Leftover Details: Interpolation, Homogeneity, Matrices
10. Dual and Proofs
11. Conclusion

EXAMPLE TWO

—

Primal PEP for function-value criterion

PEP for $f(x_1) - f_\star \leq \tau \|x_0 - x_\star\|^2$

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&= \sup_{f,\, x_0, x_1, x_\star, g_0, g_1, f_0, f_1, f_\star} f_1 - f_\star \quad , \text{ s.t. } f \in \mathcal{F}_{0,L}, x_1 = x_0 - \tfrac{1}{L} g_0, \nabla f(x_\star) = 0 \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad g_0 = \nabla f(x_0),\ g_1 = \nabla f(x_1),\ f_i = f(x_i),\ f_\star = f(x_\star) \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \|x_0 - x_\star\|^2 = 1, \\
&= \sup_{x_0, x_1, x_\star, g_0, g_1, f_0, f_1} f_1 - f_\star \\
&\qquad \text{s.t.} \quad x_1 = x_0 - \tfrac{1}{L} g_0,\ \|x_0 - x_\star\|^2 = 1 \\
&\qquad\qquad \exists f \in \mathcal{F}_{0,L}:\ g_0 = \nabla f(x_0),\ g_1 = \nabla f(x_1), \\
&\qquad\qquad f_0 = f(x_0),\ f_1 = f(x_1),\ f_\star = f(x_\star),\ \nabla f(x_\star) = 0
\end{aligned}$$

PEP for $f(x_1) - f_\star$: interpolation and SDP lifting

Three points: x_0, $x_1 = x_0 - \frac{1}{L} g_0$, $x_\star$ ($g_\star = 0$). Interpolation on $\mathcal{F}_{0,L}$: $\forall\, i, j$:

$$f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \frac{1}{2L} \|g_i - g_j\|^2.$$

PEP for $f(x_1) - f_\star$: interpolation and SDP lifting

Three points: $x_0,\ x_1 = x_0 - \frac{1}{L} g_0,\ x_\star\ (g_\star = 0)$. Interpolation on $\mathcal{F}_{0,L}$: $\forall\, i, j$:

$$f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \tfrac{1}{2L} \|g_i - g_j\|^2.$$

SDP lifting. Gram matrix $G \succcurlyeq 0$ and function-value vector F:

$$G = \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix} \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix}^\top \succcurlyeq 0,\ G \in \mathbb{R}^{4\times 4}, \qquad F = \begin{pmatrix} f_0 \\ f_1 \\ f_\star \end{pmatrix} \in \mathbb{R}^3.$$

Each condition: $\mathrm{Tr}(G\, M_{ij}) + F^\top v_{ij} \geq 0$.

PEP for $f(x_1) - f_\star$: interpolation and SDP lifting

Three points: $x_0,\ x_1 = x_0 - \frac{1}{L} g_0,\ x_\star\ (g_\star = 0)$. Interpolation on $\mathcal{F}_{0,L}$: $\forall\, i, j$:

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$$\tau^* = \sup_{\substack{0 \preccurlyeq G,\ F \in \mathbb{R}^3 \\ \mathrm{Tr}(G\, M_{\text{init}}) = 1 \\ \mathrm{Tr}(G\, C_{\text{alg}}) = 0 \\ \mathrm{Tr}(G\, M_{ij}) + F^\top v_{ij} \geq 0}} F^\top v_{\text{perf}}$$

PEP for $f(x_1) - f_\star$: interpolation and SDP lifting

Three points: x_0, $x_1 = x_0 - \frac{1}{L} g_0$, $x_\star$ ($g_\star = 0$). Interpolation on $\mathcal{F}_{0,L}$: $\forall\, i, j$:

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(i,j)	Condition
$(\star, 0)$	$f_\star \geq f_0 + \langle g_0, x_\star - x_0 \rangle + \frac{\|g_0\|^2}{2L}$
$(0, \star)$	$f_0 \geq f_\star + \frac{\|g_0\|^2}{2L}$
$(\star, 1)$	$f_\star \geq f_1 + \langle g_1, x_\star - x_1 \rangle + \frac{\|g_1\|^2}{2L}$
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$(1, 0)$	$f_1 \geq f_0 + \langle g_0, x_1 - x_0 \rangle + \frac{\|g_1 - g_0\|^2}{2L}$

$(\star, 0)$: $\mathrm{Tr}(G\, M_{\star 0}) + F^\top v_{\star 0} \geq 0$

$$M_{\star 0} = \begin{pmatrix} 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2L} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$v_{\star 0} = (-1, 0, +1)^\top$$

PEP for $f(x_1) - f_\star$: interpolation and SDP lifting

Three points: x_0, $x_1 = x_0 - \frac{1}{L} g_0$, $x_\star$ ($g_\star = 0$). Interpolation on $\mathcal{F}_{0,L}$: $\forall\, i, j$:

$$f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \tfrac{1}{2L} \|g_i - g_j\|^2.$$

SDP lifting. Gram matrix $G \succcurlyeq 0$ and function-value vector F:

$$G = \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix} \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix}^\top \succcurlyeq 0, \quad G \in \mathbb{R}^{4\times 4}, \qquad F = \begin{pmatrix} f_0 \\ f_1 \\ f_\star \end{pmatrix} \in \mathbb{R}^3.$$

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$(\star, 0)$	$f_\star \geq f_0 + \langle g_0, x_\star - x_0 \rangle + \frac{\|g_0\|^2}{2L}$
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$(1, 0)$	$f_1 \geq f_0 + \langle g_0, x_1 - x_0 \rangle + \frac{\|g_1 - g_0\|^2}{2L}$

$(0, \star)$: $\mathrm{Tr}(G\, M_{0\star}) + F^\top v_{0\star} \geq 0$

$$M_{0\star} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2L} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$v_{0\star} = (+1, 0, -1)^\top$$

PEP for $f(x_1) - f_\star$: interpolation and SDP lifting

Three points: x_0, $x_1 = x_0 - \frac{1}{L} g_0$, $x_\star$ ($g_\star = 0$). Interpolation on $\mathcal{F}_{0,L}$: $\forall\, i, j$:

$$f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \tfrac{1}{2L} \|g_i - g_j\|^2.$$

SDP lifting. Gram matrix $G \succcurlyeq 0$ and function-value vector F:

$$G = \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix} \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix}^\top \succcurlyeq 0, \quad G \in \mathbb{R}^{4\times 4}, \qquad F = \begin{pmatrix} f_0 \\ f_1 \\ f_\star \end{pmatrix} \in \mathbb{R}^3.$$

Each condition: $\mathrm{Tr}(G\, M_{ij}) + F^\top v_{ij} \geq 0$.

$$\tau^* = \sup_{\substack{0 \preccurlyeq G,\ F \in \mathbb{R}^3 \\ \mathrm{Tr}(G\, M_{\text{init}}) = 1 \\ \mathrm{Tr}(G\, C_{\text{alg}}) = 0 \\ \mathrm{Tr}(G\, M_{ij}) + F^\top v_{ij} \geq 0}} F^\top v_{\text{perf}}$$

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$(1, 0)$	$f_1 \geq f_0 + \langle g_0, x_1 - x_0 \rangle + \frac{\|g_1 - g_0\|^2}{2L}$

$(\star, 1)$: $\mathrm{Tr}(G\, M_{\star 1}) + F^\top v_{\star 1} \geq 0$

$$M_{\star 1} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & \frac{1}{2L} \end{pmatrix}$$

$$v_{\star 1} = (0, -1, +1)^\top$$

PEP for $f(x_1) - f_\star$: interpolation and SDP lifting

Three points: x_0, $x_1 = x_0 - \frac{1}{L} g_0$, $x_\star$ ($g_\star = 0$). Interpolation on $\mathcal{F}_{0,L}$: $\forall\, i, j$:

$$f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \tfrac{1}{2L} \|g_i - g_j\|^2.$$

SDP lifting. Gram matrix $G \succcurlyeq 0$ and function-value vector F:

$$G = \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix} \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix}^\top \succcurlyeq 0, \quad G \in \mathbb{R}^{4\times 4}, \qquad F = \begin{pmatrix} f_0 \\ f_1 \\ f_\star \end{pmatrix} \in \mathbb{R}^3.$$

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(i,j)	Condition
$(\star, 0)$	$f_\star \geq f_0 + \langle g_0, x_\star - x_0 \rangle + \frac{\|g_0\|^2}{2L}$
$(0, \star)$	$f_0 \geq f_\star + \frac{\|g_0\|^2}{2L}$
$(\star, 1)$	$f_\star \geq f_1 + \langle g_1, x_\star - x_1 \rangle + \frac{\|g_1\|^2}{2L}$
$(1, \star)$	$f_1 \geq f_\star + \frac{\|g_1\|^2}{2L}$
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$(1, 0)$	$f_1 \geq f_0 + \langle g_0, x_1 - x_0 \rangle + \frac{\|g_1 - g_0\|^2}{2L}$

$(1, \star)$: $\mathrm{Tr}(G\, M_{1\star}) + F^\top v_{1\star} \geq 0$

$$M_{1\star} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2L} \end{pmatrix}$$

$$v_{1\star} = (0, +1, -1)^\top$$

PEP for $f(x_1) - f_\star$: interpolation and SDP lifting

Three points: x_0, $x_1 = x_0 - \frac{1}{L} g_0$, $x_\star$ ($g_\star = 0$). Interpolation on $\mathcal{F}_{0,L}$: $\forall\, i, j$:

$$f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \tfrac{1}{2L} \|g_i - g_j\|^2.$$

SDP lifting. Gram matrix $G \succcurlyeq 0$ and function-value vector F:

$$G = \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix} \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix}^\top \succcurlyeq 0, \quad G \in \mathbb{R}^{4\times 4}, \qquad F = \begin{pmatrix} f_0 \\ f_1 \\ f_\star \end{pmatrix} \in \mathbb{R}^3.$$

Each condition: $\mathrm{Tr}(G\, M_{ij}) + F^\top v_{ij} \geq 0$.

$$\tau^* = \sup_{\substack{0 \preccurlyeq G,\ F \in \mathbb{R}^3 \\ \mathrm{Tr}(G\, M_{\text{init}}) = 1 \\ \mathrm{Tr}(G\, C_{\text{alg}}) = 0 \\ \mathrm{Tr}(G\, M_{ij}) + F^\top v_{ij} \geq 0}} F^\top v_{\text{perf}}$$

(i,j)	Condition
$(\star, 0)$	$f_\star \geq f_0 + \langle g_0, x_\star - x_0 \rangle + \frac{\|g_0\|^2}{2L}$
$(0, \star)$	$f_0 \geq f_\star + \frac{\|g_0\|^2}{2L}$
$(\star, 1)$	$f_\star \geq f_1 + \langle g_1, x_\star - x_1 \rangle + \frac{\|g_1\|^2}{2L}$
$(1, \star)$	$f_1 \geq f_\star + \frac{\|g_1\|^2}{2L}$
$(0, 1)$	$f_0 \geq f_1 + \langle g_1, x_0 - x_1 \rangle + \frac{\|g_0 - g_1\|^2}{2L}$
$(1, 0)$	$f_1 \geq f_0 + \langle g_0, x_1 - x_0 \rangle + \frac{\|g_1 - g_0\|^2}{2L}$

$(0,1)$: $\mathrm{Tr}(G\, M_{01}) + F^\top v_{01} \geq 0$

$$M_{01} = \begin{pmatrix} 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & +\frac{1}{2} \\ 0 & 0 & -\frac{1}{2L} & +\frac{1}{2L} \\ -\frac{1}{2} & +\frac{1}{2} & +\frac{1}{2L} & -\frac{1}{2L} \end{pmatrix}$$

$$v_{01} = (+1, -1, 0)^\top$$

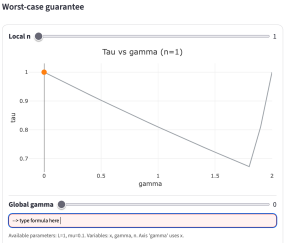
Interactive Session #2

Exploring PEP numerically with the Streamlit UI

Activity: GD on L-smooth convex functions — what is the worst-case rate for $f(x_1) - f_\star$?

1. Go to `pepitui-test.streamlit.app`
2. Select **Gradient Descent** (preimplemented) and function class $\mathcal{F}_{0,L}$ (smooth, $\mu = 0$)
3. Set $L = 1$, vary the step size γ
4. Performance metric: $f(x_1) - f_\star$ versus $\|x_0 - x_\star\|^2$
5. Run `plot` and observe the worst-case value τ^*

$\Rightarrow$ **Try to guess the formula for** τ^* as a function of L and γ.



Outline

1. Introduction & Motivation
2. Reminders: Gradient Descent and Convergence Proof
3. Performance Estimation – Squared Distance
4. Performance Estimation – Function Value
5. **The Generic PEP Framework & PEPit**
6. Minimal Working Code in PEPit
7. Primal Feasible Points as Counter-Examples
8. Hands-on: Counter-Examples & Dimension Reduction
9. Technical Leftover Details: Interpolation, Homogeneity, Matrices
10. Dual and Proofs
11. Conclusion

The Generic PEP Framework

Example	$\forall f \in \mathcal{F}_{\mu,L}$	for $x_1 = x_0 - L^{-1}\nabla f(x_0)$	then	$\|x_1 - x_\star\|^2$	$\leq \tau \|x_0 - x_\star\|^2$
Generically	$\forall f \in \mathcal{F}$	for $x_T = \mathcal{A}(x_0, (\nabla f(x_t))_{t<T})$	then	$\mathrm{Perf}(x_T)$	$\leq \tau\, \mathrm{Init}(x_0)$
$\rightarrow$	Functional class	Algorithm		Worst-case guarantee	

The Generic PEP Framework

Example	$\forall f \in \mathcal{F}_{\mu,L}$	for $x_1 = x_0 - L^{-1}\nabla f(x_0)$	then	$\|x_1 - x_\star\|^2$	$\leq \tau \|x_0 - x_\star\|^2$
Generically	$\forall f \in \mathcal{F}$	for $x_T = \mathcal{A}(x_0, (\nabla f(x_t))_{t<T})$	then	$\mathrm{Perf}(x_T)$	$\leq \tau\, \mathrm{Init}(x_0)$
$\rightarrow$	Functional class	Algorithm		Worst-case guarantee	

Equivalently:

$$\tau^* = \max_{f,\, x_0,\, x_T} \frac{\mathrm{Perf}(x_T)}{\mathrm{Init}(x_0)}$$

s.t. $f \in \mathcal{F}$, $x_T = \mathcal{A}(x_0, (\nabla f(x_t))_{t<T})$

$\rightarrow$ Non-convex problem

$\rightarrow$ Infinite-dimensional over f

The Generic PEP Framework

Example	$\forall f \in \mathcal{F}_{\mu,L}$	for $x_1 = x_0 - L^{-1}\nabla f(x_0)$	then	$\|x_1 - x_\star\|^2$	$\leq \tau \|x_0 - x_\star\|^2$
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$\rightsquigarrow$ **Sampling, Interpolation, SDP lifting**

$$\tau^* = \sup_{\substack{0 \preccurlyeq G,\ F \in \mathbb{R}^p \\ \mathrm{Tr}(G\, M_{\mathrm{init}}) + F^\top v_{init} = 1 \\ \forall k,\ \mathrm{Tr}(G\, M_k) + F^\top v_k = 0}} \mathrm{Tr}(G\, M_{\mathrm{perf}}) + F^\top v_{\mathrm{perf}}$$

The Generic PEP Framework

Example	$\forall f \in \mathcal{F}_{\mu,L}$	for $x_1 = x_0 - L^{-1}\nabla f(x_0)$	then	$\|x_1 - x_\star\|^2$	$\leq \tau \|x_0 - x_\star\|^2$
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Equivalently:

$$\tau^* = \max_{f,\, x_0,\, x_T} \frac{\mathrm{Perf}(x_T)}{\mathrm{Init}(x_0)}$$

s.t. $f \in \mathcal{F}$, $x_T = \mathcal{A}(x_0, (\nabla f(x_t))_{t<T})$

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$$\tau^* = \sup_{\substack{0 \preccurlyeq G,\ F \in \mathbb{R}^p \\ \mathrm{Tr}(G\, M_{\mathrm{init}}) + F^\top v_{init} = 1 \\ \forall k,\ \mathrm{Tr}(G\, M_k) + F^\top v_k = 0}} \mathrm{Tr}(G\, M_{\mathrm{perf}}) + F^\top v_{\mathrm{perf}}$$

$\forall k$ aggregates algorithm & class (interpolation) constraints:

$$\begin{cases} \forall i,\ \mathrm{Tr}(G\, M_i^{\mathrm{alg}}) = 0 \\ \forall j,\ \mathrm{Tr}(G\, M_j^{\mathrm{class}}) + F^\top v_k \geq 0 \end{cases}$$

The Generic PEP Framework

Example	$\forall f \in \mathcal{F}_{\mu,L}$	for $x_1 = x_0 - L^{-1}\nabla f(x_0)$	then	$\|x_1 - x_\star\|^2$	$\leq \tau \|x_0 - x_\star\|^2$
Generically	$\forall f \in \mathcal{F}$	for $x_T = \mathcal{A}(x_0, (\nabla f(x_t))_{t<T})$	then	$\mathrm{Perf}(x_T)$	$\leq \tau\, \mathrm{Init}(x_0)$
$\rightarrow$	Functional class	Algorithm		Worst-case guarantee	

Equivalently:

$$\tau^* = \max_{f,\, x_0,\, x_T} \frac{\mathrm{Perf}(x_T)}{\mathrm{Init}(x_0)}$$

s.t. $f \in \mathcal{F}$, $x_T = \mathcal{A}(x_0, (\nabla f(x_t))_{t<T})$

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$\forall k$ aggregates algorithm & class (interpolation) constraints:

$$\begin{cases} \forall i,\ \mathrm{Tr}(G\, M_i^{\mathrm{alg}}) = 0 \\ \forall j,\ \mathrm{Tr}(G\, M_j^{\mathrm{class}}) + F^\top v_k \geq 0 \end{cases}$$

Worst-case guarantees with PEP

$\rightarrow$ Finding a tight WCG = solving a convex SDP

♡ Automatic numerical WCG $\rightarrow$ design & proof-checking

♡ Works for many algorithms & function classes

PEPit: computer-assisted worst-case analysis in Python

Goals of PEPit:

- Avoids manual SDP modeling steps
- Collaborative, easy-to-use methodology
- Code as close as possible to mathematical specification

Supported setups:

- Algorithms: Gradient, Prox, Inexact-Gradient, Line-search, ...
- Function classes: smooth, strongly convex, quadratically upper bounded, ...
- Performance metrics: $f(x_T) - f_\star$, $\|x_T - x_\star\|^2$, $\|\nabla f(x_T)\|^2$, ...

- **80+ documented examples**

- 1. Unconstrained convex minimization
  - 1.1. Gradient descent
  - 1.2. Subgradient method
  - 1.3. Subgradient method under restricted secant inequality and error bound
  - 1.4. Gradient descent with exact line search
  - 1.5. Conjugate gradient
  - 1.6. Heavy Ball momentum
  - 1.7. Accelerated gradient for convex objective
  - 1.8. Accelerated gradient for strongly convex objective
  - 1.9. Optimized gradient
  - 1.10. Optimized gradient for gradient
  - 1.11. Robust momentum
  - 1.12. Triple momentum
  - 1.13. Information theoretic exact method
  - 1.14. Proximal point
  - 1.15. Accelerated proximal point
  - 1.16. Inexact gradient descent
  - 1.17. Inexact gradient descent with exact line search
  - 1.18. Inexact accelerated gradient
  - 1.19. Epsilon-subgradient method
  - 1.20. Gradient descent for quadratically upper bounded convex objective
  - 1.21. Gradient descent with decreasing step sizes for quadratically upper bounded convex objective
  - 1.22. Conjugate gradient for quadratically upper bounded convex objective
  - 1.23. Heavy Ball momentum for quadratically upper bounded convex objective
- 2. Composite convex minimization
  - 2.1. Proximal gradient
  - 2.2. Accelerated proximal gradient
  - 2.3. Bregman proximal point
  - 2.4. Douglas Rachford splitting
  - 2.5. Douglas Rachford splitting contraction
  - 2.6. Accelerated Douglas Rachford splitting
  - 2.7. Frank Wolfe
  - 2.8. Improved interior method
  - 2.9. No Lips in function value
  - 2.10. No Lips in Bregman divergence
  - 2.11. Three operator splitting
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  - 3.3. No Lips 2
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  - 5.5. Three operator splitting
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  - 8.3. Relatively inexact proximal point
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  - 9.2. Polyak steps in function value
- 10. Low dimensional worst-cases scenarios
  - 10.1. Inexact gradient
  - 10.2. Non-convex gradient descent

- Complete docs, active community

What can we do with PEP?

Performance metrics: any linear combination of $f(x_T) - f_\star$, $\|x_T - x_\star\|^2$, $\|\nabla f(x_T)\|^2$ at various timesteps.

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For *all* of these, the worst-case rate is the solution of an SDP.

PEPit solves it in a few lines of Python.

EXAMPLES

—

Beyond gradient descent: more oracles in PEP

Example 1: Proximal Step to Minimize $f \in \mathcal{F}_{\mu,L}$

Algorithm. $x_{k+1} = \mathrm{prox}_{\gamma f}(x_k) := \arg\min_x \left\{ f(x) + \frac{1}{2\gamma}\|x - x_k\|^2 \right\}$.

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In PEP. Sample two triplets:

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i.e. $\mathrm{Tr}(G\, C_{\mathrm{alg}}) = 0$ (linear in G).

Class constraint:

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💡 - The implicit equation

$$x_{k+1} = x_k - \gamma \nabla f(x_{k+1})$$

is *not solved explicitly*: it is encoded as an algorithm constraint (+ interpolation conditions)

No closed form for $\mathrm{prox}_{\gamma f}$ is needed.

Example 2: Projection onto a Convex Set

Algorithm. Projected gradient descent:

$$x_{k+1} = \Pi_C(x_k - \gamma \nabla f(x_k)), \qquad \Pi_C(y) = \arg\min_{x \in C} \|x - y\|.$$

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Connection to PEP. The indicator function $\iota_C(x) = 0$ if $x \in C$, $+\infty$ otherwise, satisfies:

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Indicator functions of bounded sets, and of balls are interpolable

$\rightarrow$ see Convex indicator, Pepit doc.

Example 3: Inexact Gradient Step

Algorithm. Instead of the exact gradient, use an inexact oracle $\tilde{g}_k$ satisfying:

$$\|\tilde{g}_k - \nabla f(x_k)\| \leq \delta \quad \text{or} \quad \|\tilde{g}_k - \nabla f(x_k)\|^2 \leq \delta^2 \|\nabla f(x_k)\|^2.$$

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In PEP:

- Sample exact gradient $g_k = \nabla f(x_k)$ (interpolation conditions for f).
- Introduce $\tilde{g}_k$ as an **additional variable**.
- The inexactness condition is a **quadratic constraint** in G:

$$\|\tilde{g}_k - g_k\|^2 \leq \delta^2 \;\Leftrightarrow\; \mathrm{Tr}(G\, M_{\text{inexact}}) \leq \delta^2.$$

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💡 inexactness = one extra quadratic constraint.

PEP & PEPit handle this via `problem.add_constraint(...)`.

→ see Inexact gradient step oracle, Pepit doc.

Example 4: Line Search

Algorithm. Exact line search along the gradient direction:

$$\gamma_k = \arg\min_{\gamma \geq 0} \; f(x_k - \gamma \nabla f(x_k)), \qquad x_{k+1} = x_k - \gamma_k \nabla f(x_k).$$

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True oracle condition. x_{k+1} is the exact minimizer of $\phi(\gamma) = f(x_k - \gamma g_k)$, so it must satisfy:

$$f(x_{k+1}) \;\leq\; f(x_k - \gamma\, g_k) \quad \forall\, \gamma \geq 0.$$

This is an **infinite** family of constraints: one for every γ.

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It's a relaxation!

Orthogonality is *necessary* for exact line search.

The feasible set of PEP is **strictly larger** than the set of true line-search instances.

$\rightarrow$ Rates are **valid** but not necessarily **tight**.

Other examples

See the documentation.

Primitive steps

- Inexact gradient step
- Exact line-search step
- Proximal step
- Inexact proximal step
- Bregman gradient step
- Bregman proximal step
- Linear optimization step
- Linearly shifted optimization step
- Epsilon-subgradient step

Figure: Primitive steps in Pepit

Outline

1. Introduction & Motivation
2. Reminders: Gradient Descent and Convergence Proof
3. Performance Estimation – Squared Distance
4. Performance Estimation – Function Value
5. The Generic PEP Framework & PEPit
6. **Minimal Working Code in PEPit**
7. Primal Feasible Points as Counter-Examples
8. Hands-on: Counter-Examples & Dimension Reduction
9. Technical Leftover Details: Interpolation, Homogeneity, Matrices
10. Dual and Proofs
11. Conclusion

Minimal PEPit Example: GD on smooth convex f – MWE

Goal: find the tightest τ such that $f(x_n) - f_\star \leq \tau \|x_0 - x_\star\|^2$ for GD with step $\gamma = 1/L$.

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```python
from PEPit import PEP
from PEPit.functions import SmoothConvexFunction

# --- Problem parameters ---
L, gamma, n = 1.0, 1.0, 3  # smoothness, step size, nb of steps

# --- Set up the PEP ---
problem = PEP()

# Declare the function class
func = problem.declare_function(SmoothConvexFunction, L=L)

# Define the optimum x_star and f_star
x_star = func.stationary_point()
f_star = func(x_star)

# Define the starting point
x0 = problem.set_initial_point()

# Initial condition: ||x0 - x_star||^2 <= 1
problem.set_initial_condition((x0 - x_star)**2 <= 1)

# Run n gradient steps
x = x0
for _ in range(n):
x = x - gamma * func.gradient(x)

# Set performance metric: f(x_n) - f_star
problem.set_performance_metric(func(x) - f_star)

# Solve the SDP
tau = problem.solve(verbose=1)
print(f"Worst-case rate: {tau:.6f}")  # should give L/(4n+2) =
    1/6 for n=1
```

```
(PEPit) Setting up the problem: performance measure is the minimum of 1 element(s)
(PEPit) Setting up the problem: Adding initial conditions and general constraints ...
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(PEPit) Primal feasibility check:
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(PEPit) Final upper bound (dual): 0.0714281380988 9529 and lower bound (primal example): 0.0714275603704179
(PEPit) Duality gap: absolute: 5.697284773814593e-07 and relative: 7.976311586548534e-06
Worst-case rate: 0.071428
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Figure: Pepit Output

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PEPit: what happens under the hood?

What you write (math):

$$\tau^* = \max_{f,\, x_0,\, x_T} \frac{\mathrm{Perf}(x_T)}{\mathrm{Init}(x_0)}$$

$$\text{s.t. } f \in \mathcal{F},\ \ x_T = \mathcal{A}(x_0, (\nabla f(x_t))_{t<T})$$

What PEPit solves (SDP):

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Output for $n = 1$, $L = 1$, $\gamma = 1/L$

- Worst-case rate: $\tau^* \approx 0.1667$

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What does the primal SDP tell us?

The PEP primal SDP (generic form):

$$\tau^* = \sup_{\substack{0 \preccurlyeq G,\ F \in \mathbb{R}^p \\ \mathrm{Tr}(G\, M_{\text{init}}) + F^\top v_{init} = 1 \\ \forall k,\ \mathrm{Tr}(G\, M_k) + F^\top v_k = 0}} \mathrm{Tr}(G\, M_{\text{perf}}) + F^\top v_{\text{perf}}$$

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$\Rightarrow$ explicit algorithm trajectory with $\mathrm{Perf}(x_T) = \mathrm{Tr}(G\, M_{\text{perf}})$; and $\mathrm{Init}(x_0) = \mathrm{Tr}(G\, M_{\text{init}})$

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Example: one step of GD on $\mathcal{F}_{0,L}$, metric $f(x_1) - f_\star \leq \tau \|x_0 - x_\star\|^2$.

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In PEPit: call `problem.solve()` then access `.eval()` to obtain values and `f.get_interpolator()` to retrieve the worst-case function values automatically.

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- In general, SDP solvers return full-rank $G^\star$ (high-dimensional).
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$\rightsquigarrow$ **Rank minimisation heuristic** (`dimension_reduction_heuristic="trace"` in PEPit):

1. First solve the SDP to get $\tau^\star$.
2. Fix the objective, add constraint $\mathrm{Tr}(G\, M_{\text{perf}}) = \tau^\star$.
3. Minimise $\mathrm{Tr}(G)$ (nuclear norm proxy for rank) subject to all constraints.
4. $\Rightarrow$ low-rank $G^\star$ $\Rightarrow$ **low-dimensional counter-example**.

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Practical outcome

Simple low-dimensional functions often result in the worst case !

— Lab 1 —

Recovering counter example functions

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Slides on hands-on

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Interpolation: definition

Definition : Interpolation on a class $\mathcal{F}$

For $\mathcal{F} \subset \mathcal{C}(\mathbb{R}^d)$, a set of triplets $(x_i, g_i, f_i)_{i \in I}$ is **interpolable** by $\mathcal{F}$, written $\mathrm{Int}(\mathcal{F}, (x_i, g_i, f_i)_i)$, if

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We derive these conditions for the three main classes below, building each one from the previous.

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Proof sketch.

($\Rightarrow$) If f is convex, the subgradient inequality holds pointwise.

($\Leftarrow$) Build $f(x) = \max_j \{ f_j + \langle g_j, x - x_j \rangle \}$ (pointwise maximum of affine functions): convex, and interpolates by construction.

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(again) Each condition is **linear in** f_i and **bilinear in** $(g_j, x_i - x_j)$
$\rightarrow$ after SDP lifting: $\mathrm{Tr}(G\, M_{ij}) + F^\top v_{ij} \geq 0$.

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$$\mathrm{Int}(\mathcal{F}_{\mu,\infty}, (x_i, g_i, f_i)_i) \iff \mathrm{Int}\left(\mathcal{F}_{0,\infty},\ \Big(x_i,\ \underbrace{g_i - \mu x_i}_{\tilde{g}_i},\ \underbrace{f_i - \tfrac{\mu}{2}\|x_i\|^2}_{\tilde{f}_i}\Big)_i\right)$$

Derivation. $f \in \mathcal{F}_{\mu,\infty}$ iff $h := f - \frac{\mu}{2}\|\cdot\|^2 \in \mathcal{F}_{0,\infty}$. Apply the convex interpolation condition to h:

$$\tilde{f}_i \geq \tilde{f}_j + \langle \tilde{g}_j, x_i - x_j \rangle \quad \Leftrightarrow \quad f_i - \tfrac{\mu}{2}\|x_i\|^2 \geq f_j - \tfrac{\mu}{2}\|x_j\|^2 + \langle g_j - \mu x_j,\ x_i - x_j \rangle.$$

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Expanding and collecting:

$$f_i \;\geq\; f_j + \langle g_j, x_i - x_j \rangle + \tfrac{\mu}{2}\|x_i - x_j\|^2.$$

(again) Each condition is quadratic in (x_i, g_i) and linear in f_i $\rightarrow$ SDP lifting works identically.

Interpolation on $\mathcal{F}_{0,L}$ (smooth convex): Fenchel duality

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Key tool: Fenchel (convex) conjugate. For $f \in \mathcal{F}_{0,L}$, the conjugate $f^*(g) = \sup_x \{\langle g, x\rangle - f(x)\}$ satisfies:

Legendre–Fenchel facts:

- $f \in \mathcal{F}_{0,L} \;\Leftrightarrow\; f^* \in \mathcal{F}_{1/L,\infty}$
- $g_i = \nabla f(x_i) \;\Leftrightarrow\; x_i = \nabla f^*(g_i)$
- $f(x_i) + f^*(g_i) = \langle g_i, x_i\rangle$ (Fenchel–Young equality)

If $f \in \mathcal{F}_{0,L}$, we define the *dual triplets*:
$f_i^* := \langle g_i, x_i\rangle - f_i$
so that $f^* \in \mathcal{F}_{1/L,\infty}$ interpolates $(g_i, x_i, f_i^*)_i$.

Interpolation on $\mathcal{F}_{0,L}$ (smooth convex): Fenchel duality

Proposition : Interpolation on $\mathcal{F}_{0,L}$

$\mathrm{Int}(\mathcal{F}_{0,L}, (x_i, g_i, f_i)_i) \iff$

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Remark: satisfying *both* sampled smoothness $\big(f_i \leq f_j + \langle g_j, x_i - x_j\rangle + \frac{L}{2}\|x_i - x_j\|^2\big)$ and sampled convexity separately is **NOT sufficient** for interpolation.

Interpolation on $\mathcal{F}_{\mu,L}$ (smooth + strongly convex)

Combining both reductions (strong convexity and Fenchel duality):

Theorem : Interpolation on $\mathcal{F}_{\mu,L}$

$\mathrm{Int}(\mathcal{F}_{\mu,L}, (x_i, g_i, f_i)_i) \iff \forall i \neq j$:

$$f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \frac{1}{2L}\|g_i - g_j\|^2 + \frac{\mu}{2(1-\mu/L)}\left\|x_i - x_j - \frac{g_i - g_j}{L}\right\|^2.$$

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Reading the condition:

$\mu = 0$: recovers the $\mathcal{F}_{0,L}$ condition (smooth convex).

$L = \infty$: recovers the $\mathcal{F}_{\mu,\infty}$ condition (strongly convex).

- Always: **one condition per ordered pair** (i, j), quadratic in (x, g), linear in f.

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(again) As always, after SDP lifting: each condition becomes $\mathrm{Tr}(G\, M_{ij}) + F^\top v_{ij} \geq 0$ for explicit matrices $M_{ij} \in \mathbb{R}^{p\times p}$ and vectors $v_{ij} \in \mathbb{R}^q$.

Homogeneity: normalising the initial condition

Proposition : Homogeneity reduction

Suppose Perf and Init are **degree-2 homogeneous** in $(x_i, g_i)_i$ and **linear in** $()f(x_i)_i)$. Then:

$$\tau^* = \sup_{\substack{f \in \mathcal{F},\ \mathcal{A} \\ \mathrm{Init}(x_0) > 0}} \frac{\mathrm{Perf}(x_T)}{\mathrm{Init}(x_0)} = \sup_{\substack{f \in \mathcal{F},\ \mathcal{A} \\ \mathrm{Init}(x_0) = 1}} \mathrm{Perf}(x_T).$$

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Fix any feasible instance (f, x_0) with $x_\star = 0$ and define the **scaled function**: $f_R(x) := R^2 f\left(\frac{x}{R}\right)$. Then:

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$\Rightarrow$ Thus the **ratio** is invariant,

$$\frac{\|x_1^R\|^2}{\|x_0^R\|^2} = \frac{R^2\|x_1\|^2}{R^2\|x_0\|^2} = \frac{\|x_1\|^2}{\|x_0\|^2},$$

so WLOG scale to $\|x_0\|^2 = 1$ by replacing $x_i \leftarrow x_i/R$, $g_i \leftarrow g_i/R$, $f_i \leftarrow f_i/R^2$.

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After lifting, if we skip the homogenization step the PEP is:

$$\tau^* = \sup_{\substack{0 \preccurlyeq G,\, F \in \mathbb{R}^p \\ \forall k,\, \mathrm{Tr}(G\, M_k) + F^\top v_k = 0}} \frac{\mathrm{Tr}(G\, M_{\mathrm{perf}}) + F^\top v_{\mathrm{perf}}}{\mathrm{Tr}(G\, M_{\mathrm{init}}) + F^\top v_{init}}$$

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$\rightarrow$ all constraints and the objective are **linear in** (G, F): (G, F) is feasible iif $(\lambda G, \lambda F)$ is.

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→ all constraints and the objective are **linear in** (G, F): (G, F) is feasible iif $(\lambda G, \lambda F)$ is.

→ For any feasible point with a given objective value, we can build another feasible point , with the same objective value and $\mathrm{Tr}(G\,M_{\mathrm{init}}) + F^\top v_{init} = 1$.

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2. Reminders: Gradient Descent and Convergence Proof
3. Performance Estimation – Squared Distance
4. Performance Estimation – Function Value
5. The Generic PEP Framework & PEPit
6. Minimal Working Code in PEPit
7. Primal Feasible Points as Counter-Examples
8. Hands-on: Counter-Examples & Dimension Reduction
9. Technical Leftover Details: Interpolation, Homogeneity, Matrices
10. **Dual and Proofs**
11. Conclusion

Primal vs Dual – two sides of the same coin

PEP-primal (find worst-case instance):

$$\tau^* = \sup_{\substack{0 \preccurlyeq G \\ \mathrm{Tr}(G\, M_{\text{init}})=1 \\ \forall k,\, \mathrm{Tr}(G\, M_k) \geq 0}} \mathrm{Tr}(G\, M_{\text{perf}})$$

- Primal feasible $G \Rightarrow$ lower bound on rate
- Primal optimal $G^\star \Rightarrow$ tight counter-example

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PEP-dual (find proof certificate):

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Strong duality (Slater's condition): $\tau^*_{\mathrm{primal}} = \tau^*_{\mathrm{dual}}$.

A rate τ is valid if and only if $\exists\, (\lambda_k) \geq 0 :\ M_{\mathrm{perf}} - \tau\, M_{\mathrm{init}} - \sum_k \lambda_k M_k \preccurlyeq 0$.

+ valeurs de fonctions

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Summary & References

What we covered:

1. GD convergence proof (co-coercivity)
2. PEP = optimization over functions $\to$ SDP
3. Step-by-step for $\|x_1 - x_\star\|^2$ and $f(x_1) - f_\star$
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Appendix